

```
In [1]: import pandas as pd
df = pd.read_excel("D:\\KSR Datavizion\\Power BI Course\\Dataset\\Sample - Superstore.xlsx")
```

```
In [2]: df.head()
```

Out[2]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	...	Postal Code	Region
0	2698	CA-2016-145317	2016-03-18	2016-03-23	Standard Class	SM-20320	Sean Miller	Home Office	United States	Jacksonville	...	32216.0	South
1	6827	CA-2018-118689	2018-10-02	2018-10-09	Standard Class	TC-20980	Tamara Chand	Corporate	United States	Lafayette	...	47905.0	Central
2	8154	CA-2019-140151	2019-03-23	2019-03-25	First Class	RB-19360	Raymond Buch	Consumer	United States	Seattle	...	98115.0	West
3	2624	CA-2019-127180	2019-10-22	2019-10-24	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	...	10024.0	East
4	4191	CA-2019-166709	2019-11-17	2019-11-22	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Newark	...	19711.0	East

5 rows × 21 columns



```
In [3]: list(df.columns)
```

```
Out[3]: ['Row ID',
'Order ID',
'Order Date',
'Ship Date',
'Ship Mode',
'Customer ID',
'Customer Name',
'Segment',
'Country/Region',
'City',
'State',
'Postal Code',
'Region',
'Product ID',
'Category',
'Sub-Category',
'Product Name',
'Sales',
'Quantity',
'Discount',
'Profit']
```

```
In [4]: df1 = df[["Region", "Country/Region", "State", "City", "Segment", "Category", "Sales", "Profit", "Quantity"]]
```

```
In [12]: df1.head()
```

```
Out[12]:
```

	Region	Country/Region	State	City	Segment	Category	Sales	Profit	Quantity
0	South	United States	Florida	Jacksonville	Home Office	Technology	22638.480	-1811.0784	6
1	Central	United States	Indiana	Lafayette	Corporate	Technology	7499.950	8399.9760	5
2	West	United States	Washington	Seattle	Consumer	Technology	3999.960	6719.9808	4
3	East	United States	New York	New York City	Home Office	Technology	11199.968	3919.9888	4
4	East	United States	Delaware	Newark	Consumer	Technology	499.970	5039.9856	3

```
In [5]: df1[df1["Region"] == "South"][["Sales"]].sum()
```

```
Out[5]: Sales    391721.905
dtype: float64
```

```
In [6]: df1[df1["Region"] == "Central"][["Sales"]].sum()
```

```
Out[6]: Sales    491239.4468
dtype: float64
```

```
In [7]: df.groupby(["Region"])[["Sales"]].sum().sort_values("Sales",ascending=False) #1 column with 1 aggreg
```

```
Out[7]:
```

	Sales
Region	
West	715457.8245
East	668781.2400
Central	491239.4468
South	391721.9050

```
In [8]: df.groupby(["Region"])[["Sales", "Profit"]].sum().sort_values("Sales",ascending=False) ##1 column wit
```

```
Out[8]:
```

	Sales	Profit
Region		
West	715457.8245	108418.4489
East	668781.2400	91522.7800
Central	491239.4468	39707.4725
South	391721.9050	46749.4303

```
In [9]: df.groupby(["Region", "Segment"])[["Sales"]].sum().sort_values("Sales", ascending=False) #1 column with
```

Out[9]:

Sales		
Region	Segment	
West	Consumer	352880.7730
East	Consumer	340908.1670
Central	Consumer	252030.9900
West	Corporate	225855.2745
East	Corporate	200409.3470
South	Consumer	195580.9710
Central	Corporate	147995.8128
West	Home Office	136721.7770
East	Home Office	127463.7260
South	Corporate	121885.9325
Central	Home Office	91212.6440
South	Home Office	74255.0015

```
In [10]: df1.groupby(["Region", "Segment"])[["Sales", "Profit"]].sum().sort_values(["Sales", "Profit"], ascending
```

Out[10]:

		Sales	Profit
Region	Segment		
West	Consumer	352880.7730	57450.6040
East	Consumer	340908.1670	41190.9843
Central	Consumer	252030.9900	8565.1581
West	Corporate	225855.2745	34437.4299
East	Corporate	200409.3470	23622.5789
South	Consumer	195580.9710	26913.5728
Central	Corporate	147995.8128	18703.9020
West	Home Office	136721.7770	16530.4150
East	Home Office	127463.7260	26709.2168
South	Corporate	121885.9325	15215.2232
Central	Home Office	91212.6440	12438.4124
South	Home Office	74255.0015	4620.6343

```
In [11]: #1 column 2 different aggregation
```

```
In [12]: df.groupby(["Region"]).aggregate({"Sales": "sum", "Profit": "mean"})
```

Out[12]:

	Sales	Profit
Region		
Central	491239.4468	17.100548
East	668781.2400	32.135808
South	391721.9050	28.857673
West	715457.8245	33.849032

```
In [13]: #2 different columns 2 different aggregation
```

```
In [14]: df1.groupby(["Region", "Segment"]).agg({"Sales": "sum", "Profit": "mean"})
```

Out[14]:

Region	Segment	Sales	Profit
Central	Consumer	252030.9900	7.072798
	Corporate	147995.8128	27.791831
	Home Office	91212.6440	28.398202
East	Consumer	340908.1670	28.040153
	Corporate	200409.3470	26.935666
	Home Office	127463.7260	53.205611
South	Consumer	195580.9710	32.116435
	Corporate	121885.9325	29.833771
	Home Office	74255.0015	16.987626
West	Consumer	352880.7730	34.360409
	Corporate	225855.2745	35.872323
	Home Office	136721.7770	28.949939

```
In [15]: df1.groupby(["Region"]).aggregate({"Sales": "sum", "Sales": "mean"}) #it is overriding the sales col
```

Out[15]:

Region	Sales
Central	211.558763
East	234.824874
South	241.803645
West	223.371160

```
In [16]: df1.groupby(["Region"]).agg({"Sales": ["sum", "mean"]}) #1 column with 2 diff agg for the same column
```

Out[16]:

Region	Sales	
	sum	mean
Central	491239.4468	211.558763
East	668781.2400	234.824874
South	391721.9050	241.803645
West	715457.8245	223.371160

```
In [17]: df.groupby(["Region", "Segment"]).agg({"Sales":["sum", "mean"], "Profit":["sum", "mean"]})
```

Out[17]:

		Sales		Profit	
		sum	mean	sum	mean
Region	Segment				
Central	Consumer	252030.9900	208.118076	8565.1581	7.072798
	Corporate	147995.8128	219.904625	18703.9020	27.791831
	Home Office	91212.6440	208.248046	12438.4124	28.398202
East	Consumer	340908.1670	232.068187	41190.9843	28.040153
	Corporate	200409.3470	228.516929	23622.5789	26.935666
	Home Office	127463.7260	253.911805	26709.2168	53.205611
South	Consumer	195580.9710	233.390180	26913.5728	32.116435
	Corporate	121885.9325	238.992025	15215.2232	29.833771
	Home Office	74255.0015	272.996329	4620.6343	16.987626
West	Consumer	352880.7730	211.053094	57450.6040	34.360409
	Corporate	225855.2745	235.265911	34437.4299	35.872323
	Home Office	136721.7770	239.442692	16530.4150	28.949939

```
In [18]: #renaming the columns with proper headers
```

```
In [19]: df1.groupby(["Region", "Segment"]).agg({"Sales":["sum", "mean"], "Profit":["sum", "mean"]}).rename(colu
```



Out[19]:

		Sales		Profit	
		Total Sales	Average Sales	Total Sales	Average Sales
Region	Segment				
Central	Consumer	252030.9900	208.118076	8565.1581	7.072798
	Corporate	147995.8128	219.904625	18703.9020	27.791831
	Home Office	91212.6440	208.248046	12438.4124	28.398202
East	Consumer	340908.1670	232.068187	41190.9843	28.040153
	Corporate	200409.3470	228.516929	23622.5789	26.935666
	Home Office	127463.7260	253.911805	26709.2168	53.205611
South	Consumer	195580.9710	233.390180	26913.5728	32.116435
	Corporate	121885.9325	238.992025	15215.2232	29.833771
	Home Office	74255.0015	272.996329	4620.6343	16.987626
West	Consumer	352880.7730	211.053094	57450.6040	34.360409
	Corporate	225855.2745	235.265911	34437.4299	35.872323
	Home Office	136721.7770	239.442692	16530.4150	28.949939

```
In [20]: #groupby with filters
```

```
In [21]: import pandas as pd
data = pd.read_csv("D:\\Data Engineering\\Python\\car_data.csv")
```

```
In [22]: data.head()
```

```
Out[22]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual

```
In [23]: data.groupby(["Year"])[["Car_Name"]].count()
```

```
Out[23]:
```

	Car_Name
Year	
2003	2
2004	1
2005	4
2006	4
2007	2
2008	7
2009	6
2010	15
2011	19
2012	23
2013	33
2014	38
2015	61
2016	50
2017	35
2018	1

```
In [24]: data.groupby(["Year"])[["Car_Name"]].count().query('Car_Name > 20')
```

```
Out[24]:
```

	Car_Name
Year	
2012	23
2013	33
2014	38
2015	61
2016	50
2017	35

```
In [25]: ta.groupby(["Year", "Fuel_Type"])[["Car_Name"]].count().query('Car_Name > 30 and Fuel_Type == "Petrol"')
```



Out[25]:

Total Car Count		
Year	Fuel_Type	
2015	Petrol	45
2016	Petrol	42
2017	Petrol	31

```
In [26]: data.head(2)
```

Out[26]:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual

```
In [36]: data.groupby(["Year", "Fuel_Type"])[["Car_Name"]]\
        .count().query('Car_Name > 30 and Fuel_Type == "Petrol"')\
        .rename(columns = {"Car_Name": "Total Car Count"})\
        .head()
```

Out[36]:

Total Car Count		
Year	Fuel_Type	
2015	Petrol	45
2016	Petrol	42
2017	Petrol	31

```
In [ ]:
```

```
In [ ]:
```